Volume 1: Base Game Engine & Real-Time Multiplayer Architecture

Comprehensive engineering contract for core game rules, turn timers, number deck arithmetic, skill mechanics, WebSocket gateway events, bot simulation heuristics, and Barnaby mascot visual state machines.

Target Stack: NestJS 11 · Socket.IO · Redis Streams · Next.js 16 (Turbopack) · **Lead:** Muhammad Sameer Ali

VOLUME OBJECTIVE:
This document specifies the exact real-time game loop and math engine for Countdown 31. It defines how turns are validated, how the 7-second countdown operates, how 2-skill loadouts execute, how bots make decisions, and how multiplayer rooms synchronize state across desktop, tablet, and mobile devices.

1. Core Mathematical Rules & Turn Validation Engine

- **Target Sum:** The goal of each match is to force the count to reach exactly **31** or force the opponent to overshoot / timeout.
- **Card Pool:** Players are presented with cards representing numbers 1, 2, and 3.
- **Turn Selection:** On their active turn, a player must select between **1 and 3 cards**. The sum of selected cards is added to the running match total.
- **Turn Timer:** Each player has a strict **7-second turn timer**. If the timer reaches 0s before move confirmation, the server auto-plays 1 card or registers a timeout elimination.
- **Victory Condition:** The player whose move pushes the count to exactly 31 wins immediately.
- **Defeat Condition:** Any move causing the count to exceed 31 results in immediate elimination.

2. Game Modes: Classic Mode vs Skill Mode

Game Mode	Gameplay Rules & Restrictions	Loadout & Cooldown Specifications
Classic Mode (Pure Counting)	Pure mathematical calculation without power-ups. Players take turns playing 1 to 3 numbers. Ideal for competitive rating and blitz tournaments.	No active skills. Standard 7s timer. Minimal HUD overlay.
Skill Mode (Arcade Powers)	Players equip a **2-Skill Loadout** before the match from the pre-game modal (SkillLoadoutModal.tsx). Skills introduce strategic disruption but are disabled when the count reaches ≥ 22 (Endgame Lock).	**Available Skills:** • **Skip:** Passes turn to rival without picking cards (Cooldown: 3 rounds). • **Reverse:** Inverts turn order in 3+ player matches (Cooldown: 2 rounds). • **Freeze:** Reduces rival's turn timer to 4 seconds (Cooldown: 4 rounds). • **Double:** Doubles the point value of played cards (Cooldown: 3 rounds).

3. Real-Time WebSocket Gateway Protocol (`/countdown`)

The backend implements a NestJS WebSocket Gateway (CountdownGateway) with Redis Adapter for multi-instance sync:

Event Name	Direction & Payload	Server Handling & State Transition
`room:join`	Client → Server { roomId: string, username: string, loadout?: string[] }	Validates room availability. Adds socket to Redis room set. Broadcasts updated player roster to all clients in room.
`match:state_sync`	Server → Client { count: number, activePlayerId: string, remainingSeconds: number, lastMove: MoveDto, status: 'waiting'|'playing'|'over' }	Full state snapshot emitted on join, reconnect, or turn completion.
`turn:move_submit`	Client → Server { cards: number[], skillUsed?: string }	Verifies active turn socket identity. Validates card values (1-3) and skill availability. Increments running count. Resets 7s timer. Emits turn result.
`player:eliminated`	Server → Client { playerId: string, reason: 'timeout'|'bust', count: number }	Broadcasts elimination. Triggers defeat audio and mascot animation on client.
`match:game_over`	Server → Client { winnerId: string, prizeUsdt: string, finalCount: 31, stats: MatchStatsDto }	Closes match, calculates USDT payouts, updates player win records in DB, emits fanfare.

4. Bot Simulation Engine & AI Decision Heuristics

When human players are waiting or practicing, the backend bot runner simulates realistic opponent play:
• **Mathematical Strategy:** AI seeks target key numbers (e.g. 27, 23, 19, 15, 11, 7, 3) to guarantee a win path.
• **Thinking Latency:** Random delay between 1.5s and 4.2s to emulate real human decision-making.
• **Mistake Rate:** Tuned by bot difficulty level (5% on Master, 20% on Standard, 45% on Casual).
• **Timeout Fallback:** In live tournaments, if a human player disconnects, an AI surrogate takes over after 1 turn timeout.

5. Mascot Animation State Machine (Barnaby Cow)

The visual mascot (BarnabyMascot.tsx) reacts dynamically to game state changes: `idle` (chewing grass), `thinking` (hand on chin when player is picking cards), `cheering` (high score move), `defeat` (gong sound + sad slump), `victory_dance` (crown animation + golden confetti).